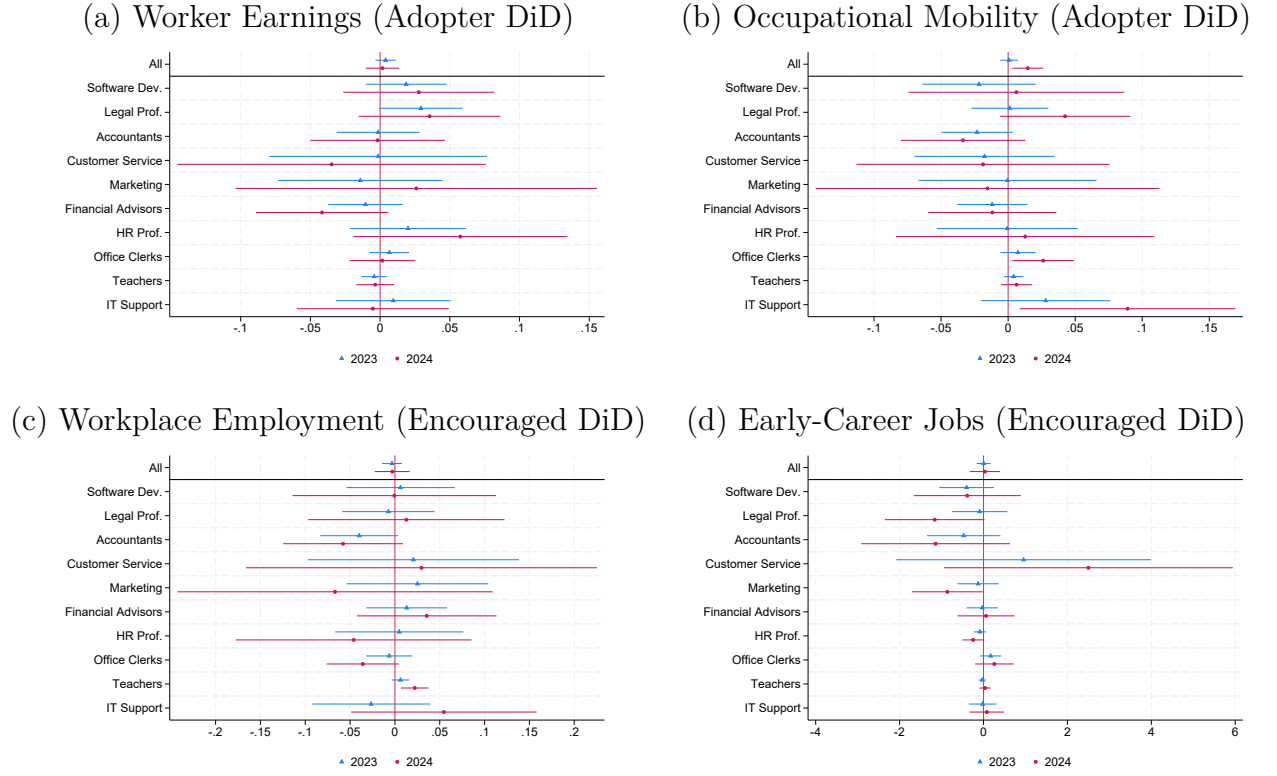


Figure 4: Labor Market Effects of AI Chatbots: Heterogeneity by Strength of Treatment



Notes: This figure shows how our difference-in-differences estimates vary with the strength of the treatment. Estimates are based on the pooled specification in Equation (2), with 95% confidence intervals shown as whiskers. Panels (a) and (b) focus on log earnings and occupational mobility for different adopters relative to non-encouraged non-adopters (Figure 2). Occupational mobility is defined as full-time-equivalent (FTE) employment in workers' latest (December 2024) occupations. *Encouraged + Training* restricts the treatment group to encouraged adopters who received employer-provided training; *Encouraged + Enterprise Chatbot*, to those whose employer deployed an enterprise chatbot; and *Encouraged + Enterprise Chatbot + Training*, to those with both. *Daily Use* refers to users reporting daily chatbot use at work. *Early Adopters* adopted ChatGPT by the time of our 2023 survey. *Large Time Savings* indicates time savings of more than 60 minutes per day of usage. *Increased Quality / Creativity* refers to users reporting improved output quality or creativity. *Increased Workload* refers to users reporting more workloads due to AI chatbots. *AI Task* restricts to workers who report new AI-related job tasks. Panels (c) and (d) show workplace outcomes (log employment and early-career employment) for workers who report encouraged usage policies relative to those without encouragement (Figure 3). *Encouraged + Training* restricts the treatment group to encouraged workers who received employer-provided training; *Encouraged + Enterprise Chatbot*, to those whose employer deployed an enterprise chatbot; and *Encouraged + Enterprise Chatbot + Training*, to those with both. *Sample:* All completed 2024 survey responses linked to registry data.

Figure 5: Labor Market Effects of AI Chatbots: Heterogeneity by Occupation



Notes: This figure shows occupation-specific effects of AI chatbot adoption. Estimates are based on the pooled specification in Equation (2), with 95% confidence intervals shown as whiskers. Panels (a) and (b) focus on log earnings and occupational mobility for adopters relative to non-encouraged non-adopters (Figure 2). Occupational mobility is defined as full-time-equivalent (FTE) employment in workers' latest (December 2024) occupations. Panels (c) and (d) show workplace outcomes (log employment and early-career employment) for workers who report encouraged usage policies relative to those without encouragement (Figure 3). *Sample:* All completed 2024 survey responses linked to registry data.

Table 1: Worker Adoption, Reported Benefits, and New Workloads by Employer Initiatives

	Adoption		Reported Benefits (Among Ever Used)				New Workloads (Among Ever Used)		New Workloads (Among Never Used)	
	Ever Used (1)	Daily Use (2)	Any Time Saving (3)	60+ min/day (4)	Quality (5)	Creativity (6)	Same Tasks (7)	New Tasks (8)	Same Tasks (9)	New Tasks (10)
Encouraged	0.350*** (0.009)	0.110*** (0.008)	0.087*** (0.011)	0.034*** (0.010)	0.076*** (0.014)	0.065*** (0.013)	0.020*** (0.006)	0.039*** (0.008)	-0.003 (0.003)	0.018** (0.009)
Enterprise Chatbot	0.163*** (0.012)	0.008 (0.007)	-0.000 (0.015)	-0.024** (0.011)	-0.033* (0.017)	-0.016 (0.016)	0.005 (0.008)	0.027*** (0.010)	0.020*** (0.006)	0.023*** (0.007)
Training	0.284*** (0.014)	0.019*** (0.007)	-0.065*** (0.018)	-0.006 (0.011)	-0.041** (0.017)	-0.031* (0.017)	0.015** (0.007)	0.078*** (0.016)	-0.002 (0.004)	0.050*** (0.016)
Encouraged $\times$ Enterprise Chatbot	-0.137*** (0.016)	0.015 (0.013)	0.009 (0.019)	0.014 (0.016)	0.045** (0.023)	0.009 (0.021)	0.009 (0.011)	0.001 (0.014)	-0.007 (0.009)	0.002 (0.015)
Encouraged $\times$ Training	-0.154*** (0.017)	-0.014 (0.014)	0.028 (0.023)	-0.013 (0.016)	0.066*** (0.024)	0.079*** (0.024)	-0.022** (0.011)	-0.035* (0.021)	0.018 (0.014)	-0.018 (0.032)
Enterprise Chatbot $\times$ Training	-0.089*** (0.023)	0.033** (0.016)	0.040 (0.032)	-0.011 (0.020)	0.095*** (0.033)	0.022 (0.031)	0.001 (0.015)	-0.052** (0.025)	0.056** (0.026)	-0.032 (0.029)
Encouraged $\times$ Enterprise Chatbot $\times$ Training	0.109*** (0.028)	0.038 (0.025)	0.023 (0.037)	0.043* (0.026)	-0.047 (0.039)	-0.020 (0.039)	0.011 (0.020)	0.042 (0.030)	-0.056* (0.033)	-0.050 (0.043)
Worker Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Occupation FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
No Initiative, Level	0.399	0.061	0.702	0.148	0.460	0.448	0.051	0.080	0.007	0.027
Within R^2	0.217	0.068	0.041	0.016	0.021	0.019	0.012	0.014	0.011	0.009
R^2	0.292	0.183	0.077	0.063	0.073	0.032	0.027	0.029	0.015	0.043
Observations	24796	24796	14790	14790	14790	14790	14790	14790	10006	10006

Notes: This table reports how workers' chatbot adoption (Columns 1–2), user-reported benefits (Columns 3–6), and new workloads from AI chatbots (Columns 7–10) vary with employer chatbot initiatives. *60+ min/day* indicates time savings of more than one hour per day of use. The regressions control for occupation fixed effects and workers' predetermined characteristics (age, gender, experience); see Appendix D.2.1 for details. *Sample:* All completed 2024 survey responses linked to registry data.

Online Appendix

Large Language Models, Small Labor Market Effects

Anders Humlum Emilie Vestergaard
University of Chicago University of Copenhagen

A	Theoretical Framework	3
A.1	Adoption and Work	3
A.2	Labor Market Outcomes	8
A.3	Technical Derivations	22
B	Data Construction	28
B.1	Sampling Protocol	28
B.2	Survey Sample	29
B.3	Classification of New Job Tasks	34
C	Institutional Setting	40
C.1	Wage Setting Systems in Denmark	40
D	Descriptive Evidence on Adoption	41
D.1	Employer Initiatives	42
D.2	Worker Adoption and Reported Effects	46
D.3	Total Time Savings	54
D.4	Comparison to the Literature	55
E	Robustness Analysis	58
E.1	Descriptive Evidence on Adoption	58
E.2	Results	78
E.3	Perceived Impacts	86
E.4	Additional Controls	90
E.5	Coworker Encouragement	93
F	Spillovers to Non-Adopters	98
G	Labor Market Trends	100
G.1	Aggregate Employment	100
G.2	By Age Groups (Brynjolfsson, Chandar and Chen, 2025)	100